

Data Preprocessing and Feature Selection

Task: Preprocess the data and apply feature selection to identify the most important features for predicting diabetes.

Dataset: Pima Indians Diabetes Database (diabetes.csv)

8 input features (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age) and target Outcome (0 = No Diabetes, 1 = Diabetes).

1. Load the dataset into a Pandas DataFrame.
2. Explore the data — shape, data types, first few rows.
3. Check for missing/invalid values (0s in Glucose, BloodPressure, SkinThickness, Insulin, BMI are actually missing data) and replace them with the column median.
4. Separate features (X) and target (y = Outcome).
5. Scale the features using StandardScaler.
6. Apply SelectKBest with f_classif to select the top 5 features.
7. Print selected feature names with their scores.
8. Print the correlation of each selected feature with Outcome.